

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE –25

TROPICAL CYCLONES

- NAMING OF CYCLONES
- THUNDERSTORMS AND TORNADOES

TROPICAL CYCLONES

NAMING OF CYCLONES



- 1** World Meteorological Organisation divided the **world oceans into basins** and assigned the **responsibility of naming the cyclones to the respective regional bodies**.
- 2** Each regional body has its own rules in naming cyclones.
- 3** In most regions **predetermined alphabetic list of alternative male and female names** are used.

TROPICAL CYCLONES

HOW A CYCLONES IN NAMED IN NORTHERN INDIAN OCEAN REGION?

KNOT:

- The Knot is a unit of speed equal to one nautical mile per hour.
- Worldwide, Knot is used by meteorology and in maritime and air navigation to calculate the speed of the corresponding wind or flight or ship.

1 The names of the cyclones in the Indian ocean are not allocated in alphabetical order. But, they are arranged by the names of the country which contributed the same.

2 It is usual practice for a storm to be named when it reaches tropical storms strength (winds of 34 knots).

TROPICAL CYCLONES

HOW A CYCLONES IN NAMED IN NORTHERN INDIAN OCEAN REGION?



- 3** The Indian Meteorological Department (IMD), which issues cyclone advises to 8 countries has list of names contributed by each of those eight countries.
- 4** Every time a cyclone occurs the name is given in the order of the names that are already submitted.
- 5** Each country gets a chance to name as a cyclone. After all the countries get their turn, the next list of names is followed.

Northern Indian Ocean Names (as of 2010)

Contributors	List 1	List 2	List 3	List 4
Bangladesh	Onil	Ogni	Nisha	Giri
India	Agni	Akash	Bijli	Jal
Maldives	Hibaru	Gonu	Aila	Keila
Mayanmar	Pyarr	Yemyin	Phyan	Thane
Oman	Baaz	Sidr	Ward	Murjan
Pakistan	Fanoos	Nargis	Laila	Nilam
Sri Lanka	Mala	Rashmi	Bandu	Mahasen
Thailand	Mukda	Khai Muk	Phet	Phailin

Contributors	List 5	List 6	List 7	List 8
Bangladesh	Helen	Chapala	Ockhi	Fani
India	Lehar	Megh	Sagar	Vayu
Maldives	Madi	Roanu	Mekunu	Hikaa
Mayanmar	Nanauk	Kyant	Daye	Kyarr
Oman	Hudhud	Nada	Luban	Maha
Pakistan	Nilofar	Vardah	Titli	Bulbul
Sri Lanka	Priya	Asiri	Gigum	Soba
Thailand	Komen	Mora	Phethai	Amphan

The RSMC tropical cyclones New Delhi will give a tropical cyclone an identification name from the above name list. The identification system covers both the Arabian Sea and the Bay of Bengal.

TROPICAL CYCLONES

WHY IS THIS SYSTEM OF UNIFORMITY DONE IN NAMING A CYCLONE IN THE REGION?

WHO COMES UP WITH NAMES OF CYCLONES?

Eight countries—India, Bangladesh, Maldives, Myanmar, Oman, Pakistan, Sri Lanka and Thailand—take turns naming cyclones in the north Indian Ocean (Arabian Sea & Bay of Bengal)



- 1 Tropical cyclones are named in this system to **provide ease of communication between the forecasters and general public** regarding forecasts and warning.
- 2 Since the storms can often last a week or even longer and more than one cyclone occurring in the same region at the same time, **names can reduce the confusion about what storm is being described.**
- 3 Naming them after **a person, flower or an animal** makes it easier for the media to report on tropical cyclones.

TROPICAL CYCLONES

UPSC 2013 GS-1 QUESTION

Q 6 The recent cyclone on the east coast of India was called “Phailin”. How are the tropical cyclones named across the world?

Q 7 Tropical cyclones are largely confined to South China Sea, Bay of Bengal and Gulf of Mexico. Why?

HINTS

1 Between 6 and 20 degree north of equator

2 Free from any cold currents

3 Sea surface temperature close to 27 °C

4 High level humidity, low vertical wind shear.

TROPICAL CYCLONES

STAGES OF DEVELOPMENT OF A TROPICAL CYCLONE

- 1** Tropical cyclones evolve through a life cycle of stages from birth to death. A tropical disturbance in time can grow to a more intense stage by attaining a specified sustained wind speed.
- 2** At first, **a tropical depression would be created**. Once a depression has been created, the amount of time it takes to achieve the next stage, **tropical storm**, can take as little as half a day to as much as a couple of days. It may not happen at all.
- 3** The same may occur for the amount of time a tropical storm needs to intensify into a tropical cyclone. Atmospheric and oceanic conditions play major roles in determining these events.
- 4** The stages of development of a tropical cyclone would be:
Tropical Depressions < Tropical Storms < Tropical Cyclones < Tornadoes

THUNDERSTORMS AND TORNADOES

1 Other severe local storms are thunderstorms and tornadoes. They are of short duration, occurring over a small area but are violent.

2 Thunderstorms are caused by intense convection on moist hot days. A thunderstorm is a well-grown cumulonimbus cloud producing thunder and lightening.

3 When the clouds extend to heights where sub-zero temperature prevails, hails are formed and they come down as hailstorm.

4 If there is insufficient moisture, a thunderstorm can generate dust-storms.

5 A thunderstorms is characterised by intense updraft of rising warm air, which cause the clouds to grow bigger and rise to greater height.

6 This intense updraft of air causes precipitation.

THUNDERSTORMS AND TORNADOES

- 7 From severe thunderstorms sometimes spiralling wind descends like a trunk of an elephant with great force, with very low pressure at the centre, causing massive destruction on its way.



- 8 Such a phenomenon is called a tornado. Tornadoes generally occur in middle latitudes. The Tornado over the sea is called water sprouts.

- 9 These violent storms are the manifestation of the atmosphere's adjustments to varying energy distribution. The potential and heat energies are converted into kinetic energy in these storms and the restless atmosphere again returns to its stable state.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9